THE $K3 \times E$ CY$_3$: UNIVERSAL κ_{BKM} AND THE GENERATOR PENTAGON

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Abstract. The CY$_3$ total space $K3 \times E$ supports a chiral algebra $\mathcal{A}_{K3\times E}$ whose Borcherds–Kac–Moody shadow satisfies $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ across the five Niemeier-lattice levels $N \in \{1, 2, 3, 4, 6\}$; levels $N \geq 7$ are outside the theorem. The six candidate routes to the $K3 \times E$ generator algebra (spacetime, worldsheet heterotic, F-theory, DT, BPS, Φ-functor) do not converge isomorphically. They form a *pentagon* of five intertwiners, with the R_2 source branch carrying the minimal rank. The pentagon stratifies by *generator rank* ρ^{R_i}. The invariant κ_{ch} is a route-independent Hodge supertrace equal to 0. The $N = 1$ coincidence $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}} + \chi(\mathcal{O}_{\mathrm{fiber}})$ fails for $N \geq 2$.

Primary theorems

- `thm:borcherds-weight-kappa-BKM-universal` ($\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ universal).
- `thm:pentagon-generator-rank-stratification` (the six routes form a generator-rank pentagon).
- $\kappa_{\mathrm{ch}}(K3 \times E) = 0$ (Hodge supertrace).
- $N = 1$ coincidence failure for $N \geq 2$.

Chapter inputs (pointers)

```
\input{../chapters/examples/k3e_cy3_programme.tex}
\input{../chapters/examples/k3e_bkm_chapter.tex}
\input{../chapters/examples/cy_c_six_routes_generator_level_platonic.tex}
\input{../chapters/examples/cy_c_pentagon_hypothesis_closures_platonic.tex}
```

Corrections

The six-route convergence statement is false as an isomorphism claim. The valid structure is the generator-rank pentagon above. The pentagon is not stratified by κ_{ch}, since the Hodge supertrace gives $\kappa_{\mathrm{ch}}(K3 \times E) = 0$ on every route.

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